Year Evaluations

Organization. Each deadline has two zero topics for training.
methods expected; they can be given at the beginning of the sequence, even if all the notions
The table shows the expected time of distribution.
and B are reserved for evaluation.

Due date	Placement benchmark	Zero subjects
SP1	S9 · week of 21 September	From S2
SP2	S18 · week of 12 October	From S14
Common duty	S28 · week of 16 November ; niche to be harmonised	From S14
SP3	S34 · week of 30 November	From S21
White 1	S45 · 14-16 January, 4 h slot to be fixed	From S36
SP4	S57 · week of 8 February	From S44
SP5	S68 · week of 22 March	From S52
White 2	S71 · April 1-3, 4 h slot to be fixed	From S59
Final white	S90 · week of May 31, 4 h slot to be fixed	From S75

DS1 · Zero 1

SUBJECT ZERO · TRAINING

Duration: 2 hours · Rating of 20 · Four independent exercises.
Authorised calculator. All answers must be justified; incomplete steps will be taken in account. Give the exact values; round the probability to 10^{-6} and other approximations to 10^{-4}, unless otherwise specified.

Name: ..... **Class:** ..... **Date:** .....

Exercise 1 · Discreet evolution — 5 points

What's next? (u_n) defined by $u_0 = 1$ and, for the integer natural n, $u_{n+1} = \frac{1}{2}u_n + 4$.

1. Calculate u_1 and u_2. (0.5 pt)

2. Demonstrate by induction that, for all $n \in \mathbb{N}$, $u_n = 8 - 7\left(\frac{1}{2}\right)^n$. (2 pt)

3. Deduce that, for the integer natural n, $1 \leq u_n < 8$. (1 pt)

4. Studying the sign of $u_{n+1} - u_n$ and deduce the direction of variation of (u_n). (1 pt)

5. One student says: "There is an integer n such that $u_n = 8$. Is this statement true? Justify. (0.5 pt)

Exercise 2 · Recurrence and bounds — 5 points

What's next? (v_n) defined by $v_0 = 0$ and, for the integer natural n, $v_{n+1} = \sqrt{72 + v_n}$.

1. Calculate v_1 and v_2. The exact values will be given. (0.5 pt)

2. Demonstrate by induction that all terms are defined and that, for all $n \in \mathbb{N}$, $0 \leq v_n \leq 9$. (1.5 pt)

3. Demonstrate by induction that, for all $n \in \mathbb{N}$, $v_{n+1} \geq v_n$. (1.5 pt)

4. Demonstrate that, for everything $n \in \mathbb{N}$, $v_n < 9$. (1 pt)

5. A property is considered $P(n)$, defined for $n \in \mathbb{N}$. It has been demonstrated $P(1)$ and, for everything $n \geq 1$, Involvement $P(n) \Rightarrow P(n+1)$. Do these two results make it possible to state that $P(0)$ Is it true? Justify. (0.5 pt)

DS1 · Zero 1 · continued

Exercise 3 · Vector calculation in a base — 5 points

On a base $(\vec{\imath},\vec{\jmath},\vec{k})$ of space, vectors are defined

$$\vec{u}=\vec{\imath}+\vec{k}, \qquad \vec{v}=\vec{\jmath}+\vec{k},$$
$$\vec{w}=4\vec{\imath}+5\vec{\jmath}+9\vec{k}, \quad \vec{z}=4\vec{\imath}+5\vec{\jmath}+10\vec{k}.$$

1. Justify that $\vec{u}$ and $\vec{v}$ are not colinear. (1 pt)

2. Write $\vec{w}$ as a linear combination of $\vec{u}$ and $\vec{v}$. (0.5 pt)

3. Vectors $\vec{u}$, $\vec{v}$ and $\vec{z}$ Are they coplanaries? (1.5 pt)

4. Express vector 5 $\vec{u}-4\vec{v}$ in base $(\vec{\imath},\vec{\jmath},\vec{k})$. (1 pt)

5. Pick a point A, then we define B, C and D by $\overrightarrow{AB}=\vec{u}$, $\overrightarrow{AC}=\vec{v}$ and $\overrightarrow{AD}=\vec{w}$. Demonstrate that A, B, C and D are coplanar. (1 pt)

Exercise 4 · Geometry of a parallelepipeda — 5 points

A parallelepipeda is considered ABCDEFGH Not flattened. E, F, G and H are the images of the A, B, C and D by vector translation $\overrightarrow{AE}$. We're laying $\vec{\imath}=\overrightarrow{AB}$, $\vec{\jmath}=\overrightarrow{AD}$ and $\vec{k}=\overrightarrow{AE}$. The points I, J and K are defined by

$$\overrightarrow{AI}=\frac{1}{11}\vec{\imath}, \qquad \overrightarrow{AJ}=\frac{1}{11}\vec{\jmath}, \qquad \overrightarrow{AK}=\frac{1}{11}\vec{k}.$$

1. Express $\overrightarrow{AG}$ and $\overrightarrow{BH}$ in base $(\vec{\imath},\vec{\jmath},\vec{k})$. (1 pt)

2. Demonstrate that the Right (IJ) and (BD) are parallel. (1 pt)

3. Express $\overrightarrow{CG}$ in the base, then determine if $\overrightarrow{AB}$ and $\overrightarrow{CG}$ are colinear. Justify. (1 pt)

4. Note O the center of the parallelepiped. Demonstrate that O belongs to the right (AG). (1 pt)

5. Demonstrate that points I, J and K are not aligned. (1 pt)

DS1 · Zero 2

SUBJECT ZERO · TRAINING

Duration: 2 hours · Rating of 20 · Four independent exercises.
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Name: ..... **Class:** ..... **Date:** .....

Exercise 1 · Discreet evolution — 5 points

What's next? (u_n) defined by $u_0 = 2$ and, for the integer natural n, $u_{n+1} = \frac{1}{5}u_n + \frac{36}{5}$.

1. Calculate u_1 and u_2. (0.5 pt)

2. Demonstrate by induction that, for all $n \in \mathbb{N}$, $u_n = 9 - 7\left(\frac{1}{5}\right)^n$. (2 pt)

3. Deduce that, for the integer natural n, $2 \leq u_n < 9$. (1 pt)

4. Studying the sign of $u_{n+1} - u_n$ and deduce the direction of variation of (u_n). (1 pt)

5. One student says: "There is an integer n such that $u_n = 9$. Is this statement true? Justify. (0.5 pt)

Exercise 2 · Recurrence and bounds — 5 points

What's next? (v_n) defined by $v_0 = 0$ and, for the integer natural n, $v_{n+1} = \sqrt{90 + v_n}$.

1. Calculate v_1 and v_2. The exact values will be given. (0.5 pt)

2. Demonstrate by induction that all terms are defined and that, for all $n \in \mathbb{N}$, $0 \leq v_n \leq 10$. (1.5 pt)

3. Demonstrate by induction that, for all $n \in \mathbb{N}$, $v_{n+1} \geq v_n$. (1.5 pt)

4. Demonstrate that, for everything $n \in \mathbb{N}$, $v_n < 10$. (1 pt)

5. A property is considered $P(n)$, defined for $n \in \mathbb{N}$. It has been demonstrated $P(1)$ and, for everything $n \geq 1$, Involvement $P(n) \Rightarrow P(n+1)$. Do these two results make it possible to state that $P(0)$ Is it true? Justify. (0.5 pt)

DS1 · Zero 2 · continued

Exercise 3 · Vector calculation in a base — 5 points

On a base $(\vec{\imath},\vec{\jmath},\vec{k})$ of space, vectors are defined

$$\vec{u}=\vec{\imath}+\vec{k}, \qquad \vec{v}=\vec{\jmath}+\vec{k},$$
$$\vec{w}=3\vec{\imath}+7\vec{\jmath}+10\vec{k}, \quad \vec{z}=3\vec{\imath}+7\vec{\jmath}+11\vec{k}.$$

1. Justify that $\vec{u}$ and $\vec{v}$ are not colinear. (1 pt)

2. Write $\vec{w}$ as a linear combination of $\vec{u}$ and $\vec{v}$. (0.5 pt)

3. Vectors $\vec{u}$, $\vec{v}$ and $\vec{z}$ Are they coplanaries? (1.5 pt)

4. Express vector 7 $\vec{u}-3\vec{v}$ in base $(\vec{\imath},\vec{\jmath},\vec{k})$. (1 pt)

5. Pick a point A, then we define B, C and D by $\overrightarrow{AB}=\vec{u}$, $\overrightarrow{AC}=\vec{v}$ and $\overrightarrow{AD}=\vec{w}$. Demonstrate that A, B, C and D are coplanar. (1 pt)

Exercise 4 · Geometry of a parallelepipeda — 5 points

A parallelepipeda is considered ABCDEFGH Not flattened. E, F, G and H are the images of the A, B, C and D by vector translation $\overrightarrow{AE}$. We're laying $\vec{\imath}=\overrightarrow{AB}$, $\vec{\jmath}=\overrightarrow{AD}$ and $\vec{k}=\overrightarrow{AE}$. The points I, J and K are defined by

$$\overrightarrow{AI}=\frac{1}{18}\vec{\imath}, \qquad \overrightarrow{AJ}=\frac{1}{18}\vec{\jmath}, \qquad \overrightarrow{AK}=\frac{1}{18}\vec{k}.$$

1. Express $\overrightarrow{AG}$ and $\overrightarrow{BH}$ in base $(\vec{\imath},\vec{\jmath},\vec{k})$. (1 pt)

2. Demonstrate that the Right (IJ) and (BD) are parallel. (1 pt)

3. Express $\overrightarrow{CG}$ in the base, then determine if $\overrightarrow{AB}$ and $\overrightarrow{CG}$ are colinear. Justify. (1 pt)

4. Note O the center of the parallelepiped. Demonstrate that O belongs to the right (AG). (1 pt)

5. Demonstrate that points I, J and K are not aligned. (1 pt)

DS2 · Zero 1

SUBJECT ZERO · TRAINING

Duration: 2 hours · Rating of 20 · Four independent exercises.
Authorised calculator. All answers must be justified; incomplete steps will be taken in account. Give the exact values; round the probability to 10^{-6} and other approximations to 10^{-4}, unless otherwise specified.

Name: ..... **Class:** ..... **Date:** .....

Exercise 1 · Calculated and justified limits — 5 points

For the integer $n \geq 1$, we define $u_n = \frac{3n^2+4n-1}{4n^2+3}$ and $v_n = \sqrt{n^2+4n} - n$.

1. Determine the limit of (u_n) detailing the transformation of his expression. (1 pt)

2. Show that, for the integer $n \geq 1$, $v_n = \frac{4}{\sqrt{1+\frac{4}{n}}+1}$. (1 pt)

3. Deduct the limit of (v_n). (1 pt)

4. Definition $w_n = 3n^2 - 4n$ for $n \geq 1$. Determine the limit of (w_n). (1 pt)

5. One student says that the difference between two sequences tending to $+\infty$ Still tending towards 0. Is it true? Justify. (1 pt)

Exercise 2 · True or false: reasoning on limits — 5 points

For each of the following statements, indicate whether it is true or false and justify the answer. An unwarranted answer does not yield any points. The limits of follow-up are taken when n tend To $+\infty$.

1. If $u_n \to 2$ and $v_n \to -5$, then $\frac{u_n}{v_n} \to -\frac{2}{5}$. (1 pt)

2. Follow-up $u_n = 2 + (-1)^n$ converging to 2. (1 pt)

3. If $u_n \to 0$ and $v_n \to +\infty$, then $u_n v_n \to 0$. (1 pt)

4. Next steps $w_n = \frac{\sqrt{n^2+3n}}{n}$ for $n \geq 1$ converging to 1. (1 pt)

5. For $u_n = \frac{2n+5}{n}$, we have $|u_n - 2| < 0.01$ as soon as $n > 500$. (1 pt)

DS2 · Zero 1 · continued

Exercise 3 · Vector calculation in a base — 5 points

On a base $(\vec{\imath},\vec{\jmath},\vec{k})$ of space, vectors are defined

$$\vec{u}=\vec{\imath}+\vec{k}, \qquad \vec{v}=\vec{\jmath}+\vec{k},$$
$$\vec{w}=2\vec{\imath}+6\vec{\jmath}+8\vec{k}, \quad \vec{z}=2\vec{\imath}+6\vec{\jmath}+9\vec{k}.$$

1. Justify that $\vec{u}$ and $\vec{v}$ are not colinear. (1 pt)

2. Write $\vec{w}$ as a linear combination of $\vec{u}$ and $\vec{v}$. (0.5 pt)

3. Vectors $\vec{u}$, $\vec{v}$ and $\vec{z}$ Are they coplanaries? (1.5 pt)

4. Express vector 6 $\vec{u}-2\vec{v}$ in base $(\vec{\imath},\vec{\jmath},\vec{k})$. (1 pt)

5. Pick a point A, then we define B, C and D by $\overrightarrow{AB}=\vec{u}$, $\overrightarrow{AC}=\vec{v}$ and $\overrightarrow{AD}=\vec{w}$. Demonstrate that A, B, C and D are coplanar. (1 pt)

Exercise 4 · Geometry of a parallelepipeda — 5 points

A parallelepipeda is considered ABCDEFGH Not flattened. E, F, G and H are the images of the A, B, C and D by vector translation $\overrightarrow{AE}$. We're laying $\vec{\imath}=\overrightarrow{AB}$, $\vec{\jmath}=\overrightarrow{AD}$ and $\vec{k}=\overrightarrow{AE}$. The points I, J and K are defined by

$$\overrightarrow{AI}=\frac{1}{18}\vec{\imath}, \qquad \overrightarrow{AJ}=\frac{1}{18}\vec{\jmath}, \qquad \overrightarrow{AK}=\frac{1}{18}\vec{k}.$$

1. Express $\overrightarrow{AG}$ and $\overrightarrow{BH}$ in base $(\vec{\imath},\vec{\jmath},\vec{k})$. (1 pt)

2. Demonstrate that the Right (IJ) and (BD) are parallel. (1 pt)

3. Study the relative position of the rights (AB) and (CG) : are they parallel, stinging or not Coplanaries? Justify. (1 pt)

4. Note O the center of the parallelepipeda. O Does it belong to the plan? (ABG) ? Justify. (1 pt)

5. Demonstrate that points I, J and K are not aligned. (1 pt)

DS2 · Zero 2

SUBJECT ZERO · TRAINING

Duration: 2 hours · Rating of 20 · Four independent exercises.
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Name: ..... **Class:** ..... **Date:** .....

Exercise 1 · Calculated and justified limits — 5 points

For the integer $n \geq 1$, we define $u_n = \frac{2n^2+6n-1}{3n^2+1}$ and $v_n = \sqrt{n^2+6n} - n$.

1. Determine the limit of (u_n) detailing the transformation of his expression. (1 pt)

2. Show that, for the integer $n \geq 1$, $v_n = \frac{6}{\sqrt{1+\frac{6}{n}}+1}$. (1 pt)

3. Deduct the limit of (v_n). (1 pt)

4. Definition $w_n = 2n^2 - 6n$ for $n \geq 1$. Determine the limit of (w_n). (1 pt)

5. One student says that the difference between two sequences tending to $+\infty$ Still tending towards 0. Is it true? Justify. (1 pt)

Exercise 2 · True or false: reasoning on limits — 5 points

For each of the following statements, indicate whether it is true or false and justify the answer. An unwarranted answer does not yield any points. The limits of follow-up are taken when n tend To $+\infty$.

1. If $u_n \to 5$ and $v_n \to -6$, then $\frac{u_n}{v_n} \to -\frac{5}{6}$. (1 pt)

2. Follow-up $u_n = 5 + (-1)^n$ converging to 5. (1 pt)

3. If $u_n \to 0$ and $v_n \to +\infty$, then $u_n v_n \to 0$. (1 pt)

4. Next steps $w_n = \frac{\sqrt{n^2+n}}{n}$ for $n \geq 1$ converging to 1. (1 pt)

5. For $u_n = \frac{5n+6}{n}$, we have $|u_n - 5| < 0.01$ as soon as $n > 600$. (1 pt)

DS2 · Zero 2 · continued

Exercise 3 · Vector calculation in a base — 5 points

On a base $(\vec{\imath},\vec{\jmath},\vec{k})$ of space, vectors are defined

$$\vec{u}=\vec{\imath}+\vec{k}, \qquad \vec{v}=\vec{\jmath}+\vec{k},$$
$$\vec{w}=4\vec{\imath}+7\vec{\jmath}+11\vec{k}, \quad \vec{z}=4\vec{\imath}+7\vec{\jmath}+12\vec{k}.$$

1. Justify that $\vec{u}$ and $\vec{v}$ are not colinear. (1 pt)

2. Write $\vec{w}$ as a linear combination of $\vec{u}$ and $\vec{v}$. (0.5 pt)

3. Vectors $\vec{u}$, $\vec{v}$ and $\vec{z}$ Are they coplanaries? (1.5 pt)

4. Express vector 7 $\vec{u}-4\vec{v}$ in base $(\vec{\imath},\vec{\jmath},\vec{k})$. (1 pt)

5. Pick a point A, then we define B, C and D by $\overrightarrow{AB}=\vec{u}$, $\overrightarrow{AC}=\vec{v}$ and $\overrightarrow{AD}=\vec{w}$. Demonstrate that A, B, C and D are coplanar. (1 pt)

Exercise 4 · Geometry of a parallelepipeda — 5 points

A parallelepipeda is considered ABCDEFGH Not flattened. E, F, G and H are the images of the A, B, C and D by vector translation $\overrightarrow{AE}$. We're laying $\vec{\imath}=\overrightarrow{AB}$, $\vec{\jmath}=\overrightarrow{AD}$ and $\vec{k}=\overrightarrow{AE}$. The points I, J and K are defined by

$$\overrightarrow{AI}=\frac{1}{8}\vec{\imath}, \qquad \overrightarrow{AJ}=\frac{1}{8}\vec{\jmath}, \qquad \overrightarrow{AK}=\frac{1}{8}\vec{k}.$$

1. Express $\overrightarrow{AG}$ and $\overrightarrow{BH}$ in base $(\vec{\imath},\vec{\jmath},\vec{k})$. (1 pt)

2. Demonstrate that the Right (IJ) and (BD) are parallel. (1 pt)

3. Study the relative position of the rights (AB) and (CG) : are they parallel, stinging or not Coplanaries? Justify. (1 pt)

4. Note O the center of the parallelepipeda. O Does it belong to the plan? (ABG) ? Justify. (1 pt)

5. Demonstrate that points I, J and K are not aligned. (1 pt)

Common duty · Zero 1

SUBJECT ZERO · TRAINING

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Name: ..... **Class:** ..... **Date:** .....

Exercise 1 · Discreet evolution — 5 points

What's next? (u_n) defined by $u_0 = 2$ and, for the integer natural n, $u_{n+1} = \frac{1}{4}u_n + 6$.

1. Calculate u_1 and u_2. (0.5 pt)

2. Demonstrate by induction that, for all $n \in \mathbb{N}$, $u_n = 8 - 6\left(\frac{1}{4}\right)^n$. (2 pt)

3. Deduce that, for the integer natural n, $2 \leq u_n < 8$. (1 pt)

4. Studying the sign of $u_{n+1} - u_n$ and deduce the direction of variation of (u_n). (1 pt)

5. One student says: "There is an integer n such that $u_n = 8$. Is this statement true? Justify. (0.5 pt)

Exercise 2 · Calculated and justified limits — 5 points

For the integer $n \geq 1$, we define $u_n = \frac{3n^2+5n-1}{4n^2+2}$ and $v_n = \sqrt{n^2+5n} - n$.

1. Determine the limit of (u_n) detailing the transformation of his expression. (1 pt)

2. Show that, for the integer $n \geq 1$, $v_n = \frac{5}{\sqrt{1+\frac{5}{n}}+1}$. (1 pt)

3. Deduct the limit of (v_n). (1 pt)

4. Definition $w_n = 3n^2 - 5n$ for $n \geq 1$. Determine the limit of (w_n). (1 pt)

5. One student says that the difference between two sequences tending to $+\infty$ Still tending towards 0. Is it true? Justify. (1 pt)

Common duty · Zero 1 · continued

Exercise 3 · Limitations and asymptotes — 5 points

The function is considered f defined by $f(x) = \frac{2x+6}{x-2}$. We note $\mathcal{C}_f$ its representative curve in a coordinate system in the plane.

1. Determine the definition set of f. Find two real A and B such as, for all $x \neq 2$, $f(x) = A + \frac{B}{x-2}$. (1 pt)

2. Determine the limits of f in $-\infty$ and in $+\infty$. (1 pt)

3. Determining the limits of f in 2^- and 2^+. (1 pt)

4. Give equations of asymptotes of $\mathcal{C}_f$. Specify the position of the curve in relation to its horizontal asymptote. (1 pt)

5. Irrespective of the previous questions, determine $\lim\limits_{x \to +\infty} x^2 e^{-2x}$ justifying. (1 pt)

Exercise 4 · Codes and selections — 5 points

A code is an ordered sequence of 4 symbols chosen from an alphabet that contains 15 symbols
The rules on repetition are specified in each question, independently of the
other questions.

1. How many codes can be formed if repetitions are allowed? (1 pt)

2. How many codes can be formed if each symbol is to be used at most once? (1 pt)

3. A symbol of the alphabet is fixed in advance. Repetitions are allowed, how many codes contain this symbol at least once? (1 pt)

4. The first symbol of the code is imposed and repetitions are prohibited.
Can we train? (1 pt)

5. Two distinct symbols of the alphabet are prohibited. Other symbols can be repeated.
How many codes can we train? (1 pt)

Common duty · Zero 2

SUBJECT ZERO · TRAINING

Duration: 2 hours · Rating of 20 · Four independent exercises.
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Name: ..... **Class:** ..... **Date:** .....

Exercise 1 · Discreet evolution — 5 points

What's next? (u_n) defined by $u_0 = 3$ and, for the integer natural n, $u_{n+1} = \frac{1}{3}u_n + \frac{20}{3}$.

1. Calculate u_1 and u_2. (0.5 pt)

2. Demonstrate by induction that, for all $n \in \mathbb{N}$, $u_n = 10 - 7\left(\frac{1}{3}\right)^n$. (2 pt)

3. Deduce that, for the integer natural n, $3 \leq u_n < 10$. (1 pt)

4. Studying the sign of $u_{n+1} - u_n$ and deduce the direction of variation of (u_n). (1 pt)

5. One student says: "There is an integer n such that $u_n = 10$. Is this statement true? Justify. (0.5 pt)

Exercise 2 · Calculated and justified limits — 5 points

For the integer $n \geq 1$, we define $u_n = \frac{2n^2+7n-1}{3n^2+3}$ and $v_n = \sqrt{n^2+7n} - n$.

1. Determine the limit of (u_n) detailing the transformation of his expression. (1 pt)

2. Show that, for the integer $n \geq 1$, $v_n = \frac{7}{\sqrt{1+\frac{7}{n}}+1}$. (1 pt)

3. Deduct the limit of (v_n). (1 pt)

4. Definition $w_n = 2n^2 - 7n$ for $n \geq 1$. Determine the limit of (w_n). (1 pt)

5. One student says that the difference between two sequences tending to $+\infty$ Still tending towards 0. Is it true? Justify. (1 pt)

Common duty · Zero 2 · continued

Exercise 3 · Limitations and asymptotes — 5 points

The function is considered f defined by $f(x) = \frac{5x+7}{x-3}$. We note $\mathcal{C}_f$ its representative curve in a coordinate system in the plane.

1. Determine the definition set of f. Find two real A and B such as, for all $x \neq 3$, $f(x) = A + \frac{B}{x-3}$. (1 pt)

2. Determine the limits of f in $-\infty$ and in $+\infty$. (1 pt)

3. Determining the limits of f in 3^- and 3^+. (1 pt)

4. Give equations of asymptotes of $\mathcal{C}_f$. Specify the position of the curve in relation to its horizontal asymptote. (1 pt)

5. Irrespective of the previous questions, determine $\lim\limits_{x \to +\infty} x^5 e^{-3x}$ justifying. (1 pt)

Exercise 4 · Codes and selections — 5 points

A code is an ordered sequence of 5 symbols chosen from an alphabet that contains 16 symbols
The rules on repetition are specified in each question, independently of the
other questions.

1. How many codes can be formed if repetitions are allowed? (1 pt)

2. How many codes can be formed if each symbol is to be used at most once? (1 pt)

3. A symbol of the alphabet is fixed in advance. Repetitions are allowed, how many codes contain this symbol at least once? (1 pt)

4. The first symbol of the code is imposed and repetitions are prohibited.
Can we train? (1 pt)

5. Two distinct symbols of the alphabet are prohibited. Other symbols can be repeated.
How many codes can we train? (1 pt)

DS3 · Zero 1

SUBJECT ZERO · TRAINING

Duration: 2 hours · Rating of 20 · Four independent exercises.
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Name: ..... **Class:** ..... **Date:** .....

Exercise 1 · Limits and asymptotes — 5 points

The function is considered f defined by $f(x) = \frac{5x+4}{x-1}$. We note $\mathcal{C}_f$ its representative curve in a coordinate system in the plane.

1. Determine the definition set of f. Find two real A and B such as, for all $x \neq 1$, $f(x) = A + \frac{B}{x-1}$. (1 pt)

2. Determine the limits of f in $-\infty$ and in $+\infty$. (1 pt)

3. Determining the limits of f in 1^- and 1^+. (1 pt)

4. Give equations of asymptotes of $\mathcal{C}_f$. Specify the position of the curve in relation to its horizontal asymptote. (1 pt)

5. Irrespective of the previous questions, determine $\lim\limits_{x \to +\infty} x^5 e^{-1x}$ justifying. (1 pt)

Exercise 2 · Variations, convexity and equation — 5 points

The function is considered f defined on $\mathbb{R}$ by $f(x) = x^3 + 4x - 5$. Note $\mathcal{C}_f$ its representative curve.

1. Calculate $f'(x)$, then study the direction of variation of f on $\mathbb{R}$. (1 pt)

2. Determine the limits of f in $-\infty$ and in $+\infty$. (0.5 pt)

3. Demonstrate that the equation $f(x) = 0$ admits a single real solution α. Give its exact value if it is integer; otherwise, frame it between two consecutive integers. (1.5 pt)

4. Study the convexity of f, then determine the co-ordinates of the deflection point of $\mathcal{C}_f$. (1 pt)

5. Determine an equation of tangent to $\mathcal{C}_f$ at the point of abscisse 0, then study the position of the curve in relation to this tangent. (1 pt)

DS3 · Zero 1 · continued

Exercise 3 · An exponential curve — 5 points

The function is considered f defined on $[0;+\infty[$ by $f(x) = (x+3)e^{-x}$. We note $\mathcal{C}_f$ its representative curve.

1. Calculate $f(0)$ and determine the limit of f in $+\infty$. (0.5 pt)
2. Calculate $f'(x)$, then draw up the table of variations of f. (1 pt)
3. Calculate $f''(x)$, then study the convexity of f. (1 pt)
4. Demonstrate that the equation $f(x) = \frac{3}{2}$ admits a single solution α in $[0;+\infty[$. (1.5 pt)
5. Determine an equation of tangent to $\mathcal{C}_f$ at the point of abscisse 0. Using convexity, to give an inequality between $f(x)$ and the affine expression of this tangent, for all $x \geq 0$. (1 pt)

Exercise 4 · Set up a group — 5 points

A group consists of 6 teachers and 10 students, all distinct.
The order of choice does not intervene.

1. How many groups of four people can be trained? (1 pt)
2. How many of these groups include exactly two teachers? (1 pt)
3. How many of these groups include at least one teacher and at least one student? (1 pt)
4. We select a group containing exactly two teachers, then we appoint from among its members a teacher spokesperson and a student rapporteur. Count the choices of the group accompanied of these two functions. (1 pt)
5. Explain why the number of ordered lists of four separate people, divided by 4 !, gives the number of groups obtained in question 1. (1 pt)

DS3 · Zero 2

SUBJECT ZERO · TRAINING

Duration: 2 hours · Rating of 20 · Four independent exercises.
Authorised calculator. All answers must be justified; incomplete steps will be taken in account. Give the exact values; round the probability to 10^{-6} and other approximations to 10^{-4}, unless otherwise specified.

Name: ..... **Class:** ..... **Date:** .....

Exercise 1 · Limits and asymptotes — 5 points

The function is considered f defined by $f(x) = \frac{4x+6}{x-2}$. We note $\mathcal{C}_f$ its representative curve in a coordinate system in the plane.

1. Determine the definition set of f. Find two real A and B such as, for all $x \neq 2$, $f(x) = A + \frac{B}{x-2}$. (1 pt)

2. Determine the limits of f in $-\infty$ and in $+\infty$. (1 pt)

3. Determining the limits of f in 2^- and 2^+. (1 pt)

4. Give equations of asymptotes of $\mathcal{C}_f$. Specify the position of the curve in relation to its horizontal asymptote. (1 pt)

5. Irrespective of the previous questions, determine $\lim_{x \to +\infty} x^4 e^{-2x}$ justifying. (1 pt)

Exercise 2 · Variations, convexity and equation — 5 points

The function is considered f defined on $\mathbb{R}$ by $f(x) = x^3 + 3x - 7$. Note $\mathcal{C}_f$ its representative curve.

1. Calculate $f'(x)$, then study the direction of variation of f on $\mathbb{R}$. (1 pt)

2. Determine the limits of f in $-\infty$ and in $+\infty$. (0.5 pt)

3. Demonstrate that the equation $f(x) = 0$ admits a single real solution α. Give its exact value if it is integer; otherwise, frame it between two consecutive integers. (1.5 pt)

4. Study the convexity of f, then determine the co-ordinates of the deflection point of $\mathcal{C}_f$. (1 pt)

5. Determine an equation of tangent to $\mathcal{C}_f$ at the point of abscisse 0, then study the position of the curve in relation to this tangent. (1 pt)

DS3 · Zero 2 · continued

Exercise 3 · An exponential curve — 5 points

The function is considered f defined on $[0;+\infty[$ by $f(x) = (x+2)e^{-x}$. We note $\mathcal{C}_f$ its representative curve.

1. Calculate $f(0)$ and determine the limit of f in $+\infty$. (0.5 pt)
2. Calculate $f'(x)$, then draw up the table of variations of f. (1 pt)
3. Calculate $f''(x)$, then study the convexity of f. (1 pt)
4. Demonstrate that the equation $f(x) = 1$ admits a single solution α in $[0;+\infty[$. (1.5 pt)
5. Determine an equation of tangent to $\mathcal{C}_f$ at the point of abscisse 0. Using convexity, to give an inequality between $f(x)$ and the affine expression of this tangent, for all $x \geq 0$. (1 pt)

Exercise 4 · Set up a group — 5 points

A group consists of 9 teachers and 11 students, all separate.
The order of choice does not intervene.

1. How many groups of four people can be trained? (1 pt)
2. How many of these groups include exactly two teachers? (1 pt)
3. How many of these groups include at least one teacher and at least one student? (1 pt)
4. We select a group containing exactly two teachers, then we appoint from among its members a teacher spokesperson and a student rapporteur. Count the choices of the group accompanied of these two functions. (1 pt)
5. Explain why the number of ordered lists of four separate people, divided by 4 !, gives the number of groups obtained in question 1. (1 pt)

White 1 · Zero 1

SUBJECT ZERO · TRAINING

Duration: 4 hours · Rating of 20 · Four independent exercises.
Authorised calculator. All answers must be justified; incomplete steps will be taken in account. Give the exact values; round the probability to 10^{-6} and other approximations to 10^{-4}, unless otherwise specified.

Name: ..... **Class:** ..... **Date:** .....

Exercise 1 · Recurrence and bounds — 5 points

What's next? (v_n) defined by $v_0 = 0$ and, for the integer natural n, $v_{n+1} = \sqrt{42 + v_n}$.

1. Calculate v_1 and v_2. The exact values will be given. (0.25 pt)

2. Demonstrate by induction that all terms are defined and that, for all $n \in \mathbb{N}$, $0 \leq v_n \leq 7$. (0.75 pt)

3. Demonstrate by induction that, for all $n \in \mathbb{N}$, $v_{n+1} \geq v_n$. (0.75 pt)

4. Demonstrate that, for everything $n \in \mathbb{N}$, $v_n < 7$. (0.5 pt)

5. Justify that (v_n) converge, then determine its limit. (0.75 pt)

6. Demonstrate that, for all $n \in \mathbb{N}$, $0 \leq 7 - v_{n+1} \leq \frac{7 - v_n}{7}$. (0.5 pt)

7. Deduct by induction an increase of $7 - v_n$ according to the n. Then determine, by comparison of whole powers, a ranking from which this increase guarantees $7 - v_n \leq 10^{-3}$. (1 pt)

8. Write a Python function without argument that returns the first integer n such as $7 - v_n \leq 10^{-3}$. Perform the function and compare the rank obtained to the appropriate rank of the previous question. Explain the possible difference. (0.5 pt)

Exercise 2 · An exponential curve — 5 points

The function is considered f defined on $[0; +\infty[$ by $f(x) = (x + 5)e^{-x}$. We note $\mathcal{C}_f$ its representative curve.

1. Calculate $f(0)$ and determine the limit of f in $+\infty$. (0.5 pt)

2. Calculate $f'(x)$, then draw up the table of variations of f. (0.5 pt)

3. Calculate $f''(x)$, then study the convexity of f. (0.5 pt)

4. Demonstrate that the equation $f(x) = \frac{5}{2}$ admits a single solution α in $[0; +\infty[$. (1 pt)

5. Determine an equation of tangent to $\mathcal{C}_f$ at the point of abscisse 0. Using convexity, to give an inequality between $f(x)$ and the affine expression of this tangent, for all $x \geq 0$. (0.5 pt)

6. Show that $0 < \alpha < 2$. Write a Python function that applies dichotomy on $[0; 2]$ and send back a management α Of a width not exceeding 10^{-3}. (0.5 pt)

7. Determine the minimum number of steps of dichotomy guaranteeing this width. (0.5 pt)

8. One student says: "Since $f(x)$ tend towards 0, the curve cuts the axis of the apses. » Refuse this statement by studying the sign of $f(x)$. (1 pt)

White 1 · Zero 1 · continued

Exercise 3 · Vector calculation in a base — 5 points

On a base $(\vec{\imath},\vec{\jmath},\vec{k})$ of space, vectors are defined

$$\vec{u}=\vec{\imath}+\vec{k}, \qquad \vec{v}=\vec{\jmath}+\vec{k},$$
$$\vec{w}=4\vec{\imath}+6\vec{\jmath}+10\vec{k}, \quad \vec{z}=4\vec{\imath}+6\vec{\jmath}+11\vec{k}.$$

1. Justify that $\vec{u}$ and $\vec{v}$ are not colinear. (0.5 pt)

2. Write $\vec{w}$ as a linear combination of $\vec{u}$ and $\vec{v}$. (0.5 pt)

3. Vectors $\vec{u}$, $\vec{v}$ and $\vec{z}$ Are they coplanaries? (0.5 pt)

4. Express vector 6 $\vec{u}-4\vec{v}$ in base $(\vec{\imath},\vec{\jmath},\vec{k})$. (1 pt)

5. Pick a point A, then we define B, C and D by $\overrightarrow{AB}=\vec{u}$, $\overrightarrow{AC}=\vec{v}$ and $\overrightarrow{AD}=\vec{w}$. Demonstrate that A, B, C and D are coplanar. (0.5 pt)

6. For questions 6 to 8, the orthonormized basis is assumed. AB, AC and BC. (0.5 pt)

7. Note I the midpoint of $[BC]$. Express $\overrightarrow{AI}$ in base $(\vec{\imath},\vec{\jmath},\vec{k})$, then determine whether the points A, I and D are aligned. (0.5 pt)

8. Definition E by $\overrightarrow{AE}=\vec{z}$. The right (DE) Is it included in the plan (ABC) ? Justify. (1 pt)

Exercise 4 · Repetition and estimation — 5 points

We repeat 19 times a two-run event called success and failure. The tests are independent and the probability of success is $p=\frac{3}{10}$ each test. X the number of successes obtained.

1. Justify that X follows a binomial distribution and specifies its parameters. (0.5 pt)

2. Calculate the probability of achieving exactly two successes. (0.5 pt)

3. Express with the help of contrary events, then calculate, the probability of obtaining at least two success. (0.5 pt)

4. Calculate the expectation and variance of X. (1 pt)

5. We now repeat the same test a number $N\geq 1$ of times, always independently. Determine the smallest integer N for which the probability of achieving at least one success is An array or loop search is permitted. (0.5 pt)

6. For an integer $j\in\{0,\ldots,19\}$, demonstrate the formula of $P(X=j)$ by counting positions possible successes. (0.5 pt)

7. A series of 19 events is accepted if its number of successes is between 1 and 3, bounds Calculate the probability that a series will be accepted. (0.5 pt)

8. Write a Python function without argument that returns the smallest integer j such that $P(X\leq j)\geq 0{,}95$. Run the function and check the minimality of the result. (1 pt)

White 1 · Zero 2

SUBJECT ZERO · TRAINING

Duration: 4 hours · Rating of 20 · Four independent exercises.
Authorised calculator. All answers must be justified; incomplete steps will be taken in account. Give the exact values; round the probability to 10^{-6} and other approximations to 10^{-4}, unless otherwise specified.

Name: ..... **Class:** ..... **Date:** .....

Exercise 1 · Recurrence and bounds — 5 points

What's next? (v_n) defined by $v_0 = 0$ and, for the integer natural n, $v_{n+1} = \sqrt{110 + v_n}$.

1. Calculate v_1 and v_2. The exact values will be given. (0.25 pt)

2. Demonstrate by induction that all terms are defined and that, for all $n \in \mathbb{N}$, $0 \leq v_n \leq 11$. (0.75 pt)

3. Demonstrate by induction that, for all $n \in \mathbb{N}$, $v_{n+1} \geq v_n$. (0.75 pt)

4. Demonstrate that, for everything $n \in \mathbb{N}$, $v_n < 11$. (0.5 pt)

5. Justify that (v_n) converge, then determine its limit. (0.75 pt)

6. Demonstrate that, for all $n \in \mathbb{N}$, $0 \leq 11 - v_{n+1} \leq \frac{11 - v_n}{11}$. (0.5 pt)

7. To deduct by induction an increase of $11 - v_n$ according to the n. Then determine, by comparison of entire powers, a ranking from which this increase guarantees $11 - v_n \leq 10^{-3}$. (1 pt)

8. Write a Python function without argument that returns the first integer n such as $11 - v_n \leq 10^{-3}$. Perform the function and compare the rank obtained to the appropriate rank of the previous question. Explain the possible difference. (0.5 pt)

Exercise 2 · An exponential curve — 5 points

The function is considered f defined on $[0; +\infty[$ by $f(x) = (x + 4)e^{-x}$. We note $\mathcal{C}_f$ its representative curve.

1. Calculate $f(0)$ and determine the limit of f in $+\infty$. (0.5 pt)

2. Calculate $f'(x)$, then draw up the table of variations of f. (0.5 pt)

3. Calculate $f''(x)$, then study the convexity of f. (0.5 pt)

4. Demonstrate that the equation $f(x) = 2$ admits a single solution α in $[0; +\infty[$. (1 pt)

5. Determine an equation of tangent to $\mathcal{C}_f$ at the point of abscisse 0. Using convexity, to give an inequality between $f(x)$ and the affine expression of this tangent, for all $x \geq 0$. (0.5 pt)

6. Show that $0 < \alpha < 2$. Write a Python function that applies dichotomy on $[0; 2]$ and send back a management α Of a width not exceeding 10^{-3}. (0.5 pt)

7. Determine the minimum number of steps of dichotomy guaranteeing this width. (0.5 pt)

8. One student says: "Since $f(x)$ tend towards 0, the curve cuts the axis of the apses. » Refuse this statement by studying the sign of $f(x)$. (1 pt)

White 1 · Zero 2 · continued

Exercise 3 · Vector calculation in a base — 5 points

On a base $(\vec{\imath},\vec{\jmath},\vec{k})$ of space, vectors are defined

$$\vec{u}=\vec{\imath}+\vec{k}, \qquad \vec{v}=\vec{\jmath}+\vec{k},$$
$$\vec{w}=3\vec{\imath}+8\vec{\jmath}+11\vec{k}, \quad \vec{z}=3\vec{\imath}+8\vec{\jmath}+12\vec{k}.$$

1. Justify that $\vec{u}$ and $\vec{v}$ are not colinear. (0.5 pt)

2. Write $\vec{w}$ as a linear combination of $\vec{u}$ and $\vec{v}$. (0.5 pt)

3. Vectors $\vec{u}$, $\vec{v}$ and $\vec{z}$ Are they coplanaries? (0.5 pt)

4. Express vector 8 $\vec{u}-3\vec{v}$ in base $(\vec{\imath},\vec{\jmath},\vec{k})$. (1 pt)

5. Pick a point A, then we define B, C and D by $\overrightarrow{AB}=\vec{u}$, $\overrightarrow{AC}=\vec{v}$ and $\overrightarrow{AD}=\vec{w}$. Demonstrate that A, B, C and D are coplanar. (0.5 pt)

6. For questions 6 to 8, the orthonormized basis is assumed. AB, AC and BC. (0.5 pt)

7. Note I the midpoint of $[BC]$. Express $\overrightarrow{AI}$ in base $(\vec{\imath},\vec{\jmath},\vec{k})$, then determine whether the points A, I and D are aligned. (0.5 pt)

8. Definition E by $\overrightarrow{AE}=\vec{z}$. The right (DE) Is it included in the plan (ABC) ? Justify. (1 pt)

Exercise 4 · Repetition and estimation — 5 points

We repeat 16 times a two-run test called success and failure. The tests are independent and the probability of success is $p=\frac{1}{10}$ each test. X the number of successes obtained.

1. Justify that X follows a binomial distribution and specifies its parameters. (0.5 pt)

2. Calculate the probability of achieving exactly two successes. (0.5 pt)

3. Express with the help of contrary events, then calculate, the probability of obtaining at least two success. (0.5 pt)

4. Calculate the expectation and variance of X. (1 pt)

5. We now repeat the same test a number $N\geq 1$ of times, always independently. Determine the smallest integer N for which the probability of achieving at least one success is An array or loop search is permitted. (0.5 pt)

6. For an integer $j\in\{0,\ldots,16\}$, demonstrate the formula of $P(X=j)$ by counting positions possible successes. (0.5 pt)

7. A series of 16 events is accepted if the number of hits is between 1 and 3 points Calculate the probability that a series will be accepted. (0.5 pt)

8. Write a Python function without argument that returns the smallest integer j such that $P(X\leq j)\geq 0{,}95$. Run the function and check the minimality of the result. (1 pt)

DS4 · Zero 1

SUBJECT ZERO · TRAINING

Duration: 2 hours · Rating of 20 · Four independent exercises.
Authorised calculator. All answers must be justified; incomplete steps will be taken in account. Give the exact values; round the probability to 10^{-6} and other approximations to 10^{-4}, unless otherwise specified.

Name: ..... **Class:** ..... **Date:** .....

Exercise 1 · A logarithmic function — 5 points

The function is considered f defined on $]0;+\infty[$ by $f(x)=\frac{3+\ln x}{x}$. We note $\mathcal{C}_f$ its representative curve.

1. Determine the limits of f in 0^+ and in $+\infty$. (1 pt)

2. Calculate $f'(x)$, then draw up the table of variations of f. (1 pt)

3. Specify the maximum value of f, then solve $f(x)=0$. (1 pt)

4. Study the convexity of f and determine the coordinates of the deflection point of $\mathcal{C}_f$. (1 pt)

5. Demonstrate that the equation $f(x)=\frac{1}{2}e^2$ admits exactly two solutions in $]0;+\infty[$. (1 pt)

Exercise 2 · Primitives and imposed condition — 5 points

The first four questions are independent. For each, specify the expression of antiderivatives. and verify the result by bypass.

1. Determine all antiderivatives on $\mathbb{R}$ of $x\mapsto 2x^2-6x+2$. (1 pt)

2. Determine the antiderivative F of $x\mapsto e^{2x}$ on $\mathbb{R}$ such that $F(0)=6$. (1 pt)

3. Determine all antiderivatives on $\mathbb{R}$ of $x\mapsto\frac{2x}{x^2+6}$. (1 pt)

4. Determine a antiderivative on $\mathbb{R}$ of $x\mapsto 2x(x^2+2)^2$. (1 pt)

5. Let F and G two antiderivatives of the same function over an interval I. Demonstrate that there is a actual constant C such that $F(x)-G(x)=C$ for every $x\in I$. (1 pt)

DS4 · Zero 1 · continued

Exercise 3 · Differential equation and evolution — 5 points

The differential equation is considered $(E) : y' = -\frac{1}{5}y + \frac{8}{5}$ on $[0; +\infty[$. In questions 2 to 5, it is note y the solution that verifies $y(0) = 13$.

1. Determine the constant solution of (E). (1 pt)

2. Determine the expression of the solution verifying the given initial condition. (1 pt)

3. Determine the limit of $y(t)$ where t tend towards $+\infty$. (1 pt)

4. Study the direction of variation in y, then determine the smallest real $t \geq 0$ as $y(t) \leq 9$. (1 pt)

5. The Euler method is applied with a step $h > 0$ and we note Y_k I-approximation $y(kh)$. Provide the value of Y_0 and to express Y_{k+1} according to the Y_k and h. (1 pt)

Exercise 4 · Projection and volume — 5 points

In an orthonormal coordinate system of origin O, we consider $A(11;0;0)$, $B(0;11;0)$ and $C(0;0;11)$. We give $\vec{n} = (1;1;1)$. The lengths, areas and volumes shall be expressed respectively in units of length, and volume.

1. Demonstrate that $\vec{n}$ is normal on the plane (ABC). (1 pt)

2. Note H the orthogonal project of O on (ABC). Determine the coordinates of H using $\overrightarrow{AH} \cdot \vec{n} = 0$. (1 pt)

3. Demonstrate that OH is the distance from O on the plan (ABC) : for everything $P \in (ABC)$, establish $OP^2 \geq OH^2$. (1 pt)

4. Calculate the triangle area ABC. (1 pt)

5. Calculate the volume of the tetrahedron $OABC$ in two ways, choosing two different bases. *(1 pt)*

DS4 · Zero 2

SUBJECT ZERO · TRAINING

Duration: 2 hours · Rating of 20 · Four independent exercises.
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Name: ..... **Class:** ..... **Date:** .....

Exercise 1 · A logarithmic function — 5 points

The function is considered f defined on $]0;+\infty[$ by $f(x)=\frac{2+\ln x}{x}$. We note $\mathcal{C}_f$ its representative curve.

1. Determine the limits of f in 0^+ and in $+\infty$. (1 pt)

2. Calculate $f'(x)$, then draw up the table of variations of f. (1 pt)

3. Specify the maximum value of f, then solve $f(x)=0$. (1 pt)

4. Study the convexity of f and determine the coordinates of the deflection point of $\mathcal{C}_f$. (1 pt)

5. Demonstrate that the equation $f(x)=\frac{1}{2}e^1$ admits exactly two solutions in $]0;+\infty[$. (1 pt)

Exercise 2 · Primitives and imposed condition — 5 points

The first four questions are independent. For each, specify the expression of antiderivatives. and verify the result by bypass.

1. Determine all antiderivatives on $\mathbb{R}$ of $x\mapsto 5x^2-7x+3$. (1 pt)

2. Determine the antiderivative F of $x\mapsto e^{5x}$ on $\mathbb{R}$ such that $F(0)=7$. (1 pt)

3. Determine all antiderivatives on $\mathbb{R}$ of $x\mapsto \frac{2x}{x^2+7}$. (1 pt)

4. Determine a antiderivative on $\mathbb{R}$ of $x\mapsto 2x(x^2+3)^5$. (1 pt)

5. Let F and G two antiderivatives of the same function over an interval I. Demonstrate that there is a actual constant C such that $F(x)-G(x)=C$ for every $x\in I$. (1 pt)

DS4 · Zero 2 · continued

Exercise 3 · Differential equation and evolution — 5 points

The differential equation is considered $(E) : y' = -\frac{1}{4}y + \frac{5}{2}$ on $[0; +\infty[$. In questions 2 to 5, it is note y the solution that verifies $y(0) = 16$.

1. Determine the constant solution of (E). (1 pt)

2. Determine the expression of the solution verifying the given initial condition. (1 pt)

3. Determine the limit of $y(t)$ where t tend towards $+\infty$. (1 pt)

4. Study the direction of variation in y, then determine the smallest real $t \geq 0$ as $y(t) \leq 11$. (1 pt)

5. The Euler method is applied with a step $h > 0$ and we note Y_k I-approximation $y(kh)$. Provide the value of Y_0 and to express Y_{k+1} according to the Y_k and h. (1 pt)

Exercise 4 · Projection and volume — 5 points

In an orthonormal coordinate system of origin O, we consider $A(7;0;0)$, $B(0;7;0)$ and $C(0;0;7)$. We give $\vec{n} = (1;1;1)$. The lengths, areas and volumes shall be expressed respectively in units of length, and volume.

1. Demonstrate that $\vec{n}$ is normal on the plane (ABC). (1 pt)

2. Note H the orthogonal project of O on (ABC). Determine the coordinates of H using $\overrightarrow{AH} \cdot \vec{n} = 0$. (1 pt)

3. Demonstrate that OH is the distance from O on the plan (ABC) : for everything $P \in (ABC)$, establish $OP^2 \geq OH^2$. (1 pt)

4. Calculate the triangle area ABC. (1 pt)

5. Calculate the volume of the tetrahedron $OABC$ in two ways, choosing two different bases. *(1 pt)*

DS5 · Zero 1

SUBJECT ZERO · TRAINING

Duration: 2 hours · Rating of 20 · Four independent exercises.
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Name: ..... **Class:** ..... **Date:** .....

Exercise 1 · Comprehensive and follow-up — 5 points

For all natural n, we're laying $I_n = \int_0^1 x^n e^{4x}\,\mathrm{d}x$. For $n = 0$, integrande is the function $x \mapsto e^{4x}$.

1. Calculate I_0. (1 pt)

2. Using a party integration, calculate I_1. (1 pt)

3. Demonstrate, by comparing the integrands, that (I_n) is positive and decreasing. (1 pt)

4. Show that $I_n \leq \frac{e^4}{n+1}$ for every $n \in \mathbb{N}$. Deduct the limit of (I_n). (1 pt)

5. By means of integration by parties, express I_{n+1} according to the I_n and n. (1 pt)

Exercise 2 · Differential equation and evolution — 5 points

The differential equation is considered $(E) : y' = -\frac{1}{3}y + \frac{8}{3}$ on $[0;+\infty[$. In questions 2 to 5, it is note y the solution that verifies $y(0) = 12$.

1. Determine the constant solution of (E). (1 pt)

2. Determine the expression of the solution verifying the given initial condition. (1 pt)

3. Determine the limit of $y(t)$ where t tend towards $+\infty$. (1 pt)

4. Study the direction of variation in y, then determine the smallest real $t \geq 0$ as $y(t) \leq 9$. (1 pt)

5. The Euler method is applied with a step $h > 0$ and we note Y_k I-approximation $y(kh)$. Provide the value of Y_0 and to express Y_{k+1} according to the Y_k and h. (1 pt)

DS5 · Zero 1 · continued

Exercise 3 · Projection and volume — 5 points

In an orthonormal coordinate system of origin O, we consider $A(9;0;0)$, $B(0;9;0)$ and $C(0;0;9)$. We give $\vec{n} = (1;1;1)$. The lengths, areas and volumes shall be expressed respectively in units of length, and volume.

1. Demonstrate that $\vec{n}$ is normal on the plane (ABC). (1 pt)

2. Note H the orthogonal project of O on (ABC). Determine the coordinates of H using $\overrightarrow{AH} \cdot \vec{n} = 0$. (1 pt)

3. Demonstrate that OH is the distance from O on the plan (ABC) : for everything $P \in (ABC)$, establish $OP^2 \geq OH^2$. (1 pt)

4. Calculate the triangle area ABC. (1 pt)

5. Calculate the volume of the tetrahedron $OABC$ in two ways, choosing two different bases. *(1 pt)*

Exercise 4 · Amount of random variables — 5 points

Set $n = 17$. Considered n independent random variables $X_1, \ldots, X_{17}$, which all follow the Bernoulli's law of parameter $p = \frac{1}{5}$. We're laying $S = \sum_{i=1}^{17} X_i$ and $G = 9S - 17$.

1. Justify the law followed by S and calculate $E(S)$. (1 pt)

2. From the variance of a Bernoulli variable and independence, calculate $V(S)$. (1 pt)

3. Calculate $E(G)$ and $V(G)$. (1 pt)

4. Installation $M = \frac{S}{17}$. Calculate $E(M)$ and $V(M)$. (1 pt)

5. For this question only, the 17 variables are replaced by 17 variables all equal to X_1. Calculate the variance of their average and compare it with that of question 4. why the results differ. (1 pt)

DS5 · Zero 2

SUBJECT ZERO · TRAINING

Duration: 2 hours · Rating of 20 · Four independent exercises.
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Name: ..... **Class:** ..... **Date:** .....

Exercise 1 · Comprehensive and follow-up — 5 points

For all natural n, we're laying $I_n = \int_0^1 x^n e^{3x}\,\mathrm{d}x$. For $n = 0$, integrande is the function $x \mapsto e^{3x}$.

1. Calculate I_0. (1 pt)

2. Using a party integration, calculate I_1. (1 pt)

3. Demonstrate, by comparing the integrands, that (I_n) is positive and decreasing. (1 pt)

4. Show that $I_n \leq \frac{e^3}{n+1}$ for every $n \in \mathbb{N}$. Deduct the limit of (I_n). (1 pt)

5. By means of integration by parties, express I_{n+1} according to the I_n and n. (1 pt)

Exercise 2 · Differential equation and evolution — 5 points

The differential equation is considered $(E) : y' = -\frac{1}{2}y + 5$ on $[0;+\infty[$. In questions 2 to 5, it is note y the solution that verifies $y(0) = 15$.

1. Determine the constant solution of (E). (1 pt)

2. Determine the expression of the solution verifying the given initial condition. (1 pt)

3. Determine the limit of $y(t)$ where t tend towards $+\infty$. (1 pt)

4. Study the direction of variation in y, then determine the smallest real $t \geq 0$ as $y(t) \leq 11$. (1 pt)

5. The Euler method is applied with a step $h > 0$ and we note Y_k I-approximation $y(kh)$. Provide the value of Y_0 and to express Y_{k+1} according to the Y_k and h. (1 pt)

DS5 · Zero 2 · continued

Exercise 3 · Projection and volume — 5 points

In an orthonormal coordinate system of origin O, we consider $A(13;0;0)$, $B(0;13;0)$ and $C(0;0;13)$. We give $\vec{n} = (1;1;1)$. The lengths, areas and volumes shall be expressed respectively in units of length, and volume.

1. Demonstrate that $\vec{n}$ is normal on the plane (ABC). (1 pt)

2. Note H the orthogonal project of O on (ABC). Determine the coordinates of H using $\overrightarrow{AH} \cdot \vec{n} = 0$. (1 pt)

3. Demonstrate that OH is the distance from O on the plan (ABC) : for everything $P \in (ABC)$, establish $OP^2 \geq OH^2$. (1 pt)

4. Calculate the triangle area ABC. (1 pt)

5. Calculate the volume of the tetrahedron $OABC$ in two ways, choosing two different bases. *(1 pt)*

Exercise 4 · Amount of random variables — 5 points

Set $n = 14$. Considered n independent random variables $X_1, \ldots, X_{14}$, which all follow the Bernoulli's law of parameter $p = \frac{1}{4}$. We're laying $S = \sum_{i=1}^{14} X_i$ and $G = 6S - 14$.

1. Justify the law followed by S and calculate $E(S)$. (1 pt)

2. From the variance of a Bernoulli variable and independence, calculate $V(S)$. (1 pt)

3. Calculate $E(G)$ and $V(G)$. (1 pt)

4. Installation $M = \frac{S}{14}$. Calculate $E(M)$ and $V(M)$. (1 pt)

5. For this question only, the 14 variables are replaced by 14 variables all equal to X_1. Calculate the variance of their average and compare it with that of question 4. why the results differ. (1 pt)

White 2 · Zero 1

SUBJECT ZERO · TRAINING

Duration: 4 hours · Rating of 20 · Four independent exercises.
Authorised calculator. All answers must be justified; incomplete steps will be taken in account. Give the exact values; round the probability to 10^{-6} and other approximations to 10^{-4}, unless otherwise specified.

Name: ..... **Class:** ..... **Date:** .....

Exercise 1 · Comprehensive and follow-up — 5 points

For all natural n, we're laying $I_n = \int_0^1 x^n e^{5x}\,\mathrm{d}x$. For $n = 0$, integrande is the function $x \mapsto e^{5x}$.

1. Calculate I_0. (0.5 pt)

2. Using a party integration, calculate I_1. (0.5 pt)

3. Demonstrate, by comparing the integrands, that (I_n) is positive and decreasing. (0.5 pt)

4. Show that $I_n \leq \frac{e^5}{n+1}$ for every $n \in \mathbb{N}$. Deduct the limit of (I_n). (1 pt)

5. By means of integration by parties, express I_{n+1} according to the I_n and n. (0.5 pt)

6. Show that $I_n \geq \frac{1}{n+1}$ for every $n \in \mathbb{N}$. (0.5 pt)

7. We share $[0;1]$ in N intervals of the same length, $N \geq 1$. Write the amounts L_N and U_N of areas of rectangles constructed with values of e^{5x} at left and right ends. Justify $L_N \leq I_0 \leq U_N$, then calculate $U_N - L_N$. (0.5 pt)

8. Determine the smallest integer N ensuring that this framework has a width of not more than 10^{-2}. Write a Python function that returns N, L_N and U_N. (1 pt)

Exercise 2 · A logarithmic function — 5 points

The function is considered f defined on $]0;+\infty[$ by $f(x) = \frac{4+\ln x}{x}$. We note $\mathcal{C}_f$ its representative curve.

1. Determine the limits of f in 0^+ and in $+\infty$. (0.5 pt)

2. Calculate $f'(x)$, then draw up the table of variations of f. (0.5 pt)

3. Specify the maximum value of f, then solve $f(x) = 0$. (0.5 pt)

4. Study the convexity of f and determine the coordinates of the deflection point of $\mathcal{C}_f$. (1 pt)

5. Demonstrate that the equation $f(x) = \frac{1}{2}e^3$ admits exactly two solutions in $]0;+\infty[$. (0.5 pt)

6. Determine a antiderivative of f on $]0;+\infty[$. (0.5 pt)

7. Calculate the average value of f on $[1;e]$. (0.5 pt)

8. Irrespective of previous issues, resolve $\ln(x-3) + \ln(x+3) = \ln(27)$after having specified the set on which this equation is defined. (1 pt)

White 2 · Zero 1 · continued

Exercise 3 · Right, plan and planned — 5 points

In an orthonormal space coordinate system, the plane is considered $\mathcal{P}$ (d) equation $x + 2y + 2z = 10$ and point $M(3;0;0)$. The right d go through M and admits as a guiding vector $\vec{n} = (1;2;2)$.

1. Provide a parametric representation of d. (0.5 pt)

2. Demonstrate that d cut $\mathcal{P}$ in one point H, then calculate the coordinates of H. (0.5 pt)

3. Justify that H is the orthogonal project of M on $\mathcal{P}$, then calculate the distance from M to $\mathcal{P}$. (0.5 pt)

4. The plan is considered $\mathcal{Q}$ (d) equation $y = z$. Demonstrate that $\mathcal{P} \cap \mathcal{Q}$ is a right and give one parametric representation. (1 pt)

5. Verify that $H \in \mathcal{Q}$. Deducing the distance from M right $\mathcal{P} \cap \mathcal{Q}$. (0.5 pt)

6. Determine a Cartesian equation of the plane parallel to $\mathcal{P}$ passing through M. (0.5 pt)

7. Determine the angle between d and the right $\mathcal{P} \cap \mathcal{Q}$. (0.5 pt)

8. Define point N by $\overrightarrow{MN} = \vec{n}$. Calculate distances from N on the plan $\mathcal{P}$ and to the right $\mathcal{P} \cap \mathcal{Q}$, Then compare them. (1 pt)

Exercise 4 · Repetition and estimation — 5 points

We repeat 20 times a two-run event, called success and failure. The tests are independent and the probability of success is $p = \frac{3}{10}$ each test. X the number of successes obtained.

1. Justify that X follows a binomial distribution and specifies its parameters. (0.5 pt)

2. Calculate the probability of achieving exactly two successes. (0.5 pt)

3. Express with the help of contrary events, then calculate, the probability of obtaining at least two success. (0.5 pt)

4. Calculate the expectation and variance of X. (1 pt)

5. We now repeat the same test a number $N \geq 1$ of times, always independently. Determine the smallest integer N for which the probability of achieving at least one success is An array or loop search is permitted. (0.5 pt)

6. For an integer $j \in \{0, \ldots, 20\}$, demonstrate the formula of $P(X = j)$ by counting positions possible successes. (0.5 pt)

7. A series of 20 events is accepted if its number of successes is between 1 and 3, bounds Calculate the probability that a series will be accepted. (0.5 pt)

8. Write a Python function without argument that returns the smallest integer j such that $P(X \leq j) \geq 0{,}95$. Run the function and check the minimality of the result. (1 pt)

White 2 · Zero 2

SUBJECT ZERO · TRAINING

Duration: 4 hours · Rating of 20 · Four independent exercises.
Authorised calculator. All answers must be justified; incomplete steps will be taken in account. Give the exact values; round the probability to 10^{-6} and other approximations to 10^{-4}, unless otherwise specified.

Name: ..... **Class:** ..... **Date:** .....

Exercise 1 · Comprehensive and follow-up — 5 points

For all natural n, we're laying $I_n = \int_0^1 x^n e^{4x}\,\mathrm{d}x$. For $n = 0$, integrande is the function $x \mapsto e^{4x}$.

1. Calculate I_0. (0.5 pt)

2. Using a party integration, calculate I_1. (0.5 pt)

3. Demonstrate, by comparing the integrands, that (I_n) is positive and decreasing. (0.5 pt)

4. Show that $I_n \leq \frac{e^4}{n+1}$ for every $n \in \mathbb{N}$. Deduct the limit of (I_n). (1 pt)

5. By means of integration by parties, express I_{n+1} according to the I_n and n. (0.5 pt)

6. Show that $I_n \geq \frac{1}{n+1}$ for every $n \in \mathbb{N}$. (0.5 pt)

7. We share $[0;1]$ in N intervals of the same length, $N \geq 1$. Write the amounts L_N and U_N of areas of rectangles constructed with values of e^{4x} at left and right ends. Justify $L_N \leq I_0 \leq U_N$, then calculate $U_N - L_N$. (0.5 pt)

8. Determine the smallest integer N ensuring that this framework has a width of not more than 10^{-2}. Write a Python function that returns N, L_N and U_N. (1 pt)

Exercise 2 · A logarithmic function — 5 points

The function is considered f defined on $]0;+\infty[$ by $f(x) = \frac{3+\ln x}{x}$. We note $\mathcal{C}_f$ its representative curve.

1. Determine the limits of f in 0^+ and in $+\infty$. (0.5 pt)

2. Calculate $f'(x)$, then draw up the table of variations of f. (0.5 pt)

3. Specify the maximum value of f, then solve $f(x) = 0$. (0.5 pt)

4. Study the convexity of f and determine the coordinates of the deflection point of $\mathcal{C}_f$. (1 pt)

5. Demonstrate that the equation $f(x) = \frac{1}{2}e^2$ admits exactly two solutions in $]0;+\infty[$. (0.5 pt)

6. Determine a antiderivative of f on $]0;+\infty[$. (0.5 pt)

7. Calculate the average value of f on $[1;e]$. (0.5 pt)

8. Irrespective of previous issues, resolve $\ln(x-1) + \ln(x+1) = \ln(63)$after having specified the set on which this equation is defined. (1 pt)

White 2 · Zero 2 · Continue

Exercise 3 · Right, plan and planned — 5 points

In an orthonormal space coordinate system, the plane is considered $\mathcal{P}$ (d) equation $x + 2y + 2z = 6$ and point $M(1;0;0)$. The right d go through M and admits as a guiding vector $\vec{n} = (1;2;2)$.

1. Provide a parametric representation of d. (0.5 pt)

2. Demonstrate that d cut $\mathcal{P}$ in one point H, then calculate the coordinates of H. (0.5 pt)

3. Justify that H is the orthogonal project of M on $\mathcal{P}$, then calculate the distance from M to $\mathcal{P}$. (0.5 pt)

4. The plan is considered $\mathcal{Q}$ (d) equation $y = z$. Demonstrate that $\mathcal{P} \cap \mathcal{Q}$ is a right and give one parametric representation. (1 pt)

5. Verify that $H \in \mathcal{Q}$. Deducing the distance from M right $\mathcal{P} \cap \mathcal{Q}$. (0.5 pt)

6. Determine a Cartesian equation of the plane parallel to $\mathcal{P}$ passing through M. (0.5 pt)

7. Determine the angle between d and the right $\mathcal{P} \cap \mathcal{Q}$. (0.5 pt)

8. Define point N by $\overrightarrow{MN} = \vec{n}$. Calculate distances from N on the plan $\mathcal{P}$ and to the right $\mathcal{P} \cap \mathcal{Q}$, Then compare them. (1 pt)

Exercise 4 · Repetition and estimation — 5 points

We repeat 17 times a two-run event called success and failure. The tests are independent and the probability of success is $p = \frac{3}{10}$ each test. X the number of successes obtained.

1. Justify that X follows a binomial distribution and specifies its parameters. (0.5 pt)

2. Calculate the probability of achieving exactly two successes. (0.5 pt)

3. Express with the help of contrary events, then calculate, the probability of obtaining at least two success. (0.5 pt)

4. Calculate the expectation and variance of X. (1 pt)

5. We now repeat the same test a number $N \geq 1$ of times, always independently. Determine the smallest integer N for which the probability of achieving at least one success is An array or loop search is permitted. (0.5 pt)

6. For an integer $j \in \{0, \ldots, 17\}$, demonstrate the formula of $P(X = j)$ by counting positions possible successes. (0.5 pt)

7. A series of 17 events is accepted if the number of hits is between 1 and 3 points Calculate the probability that a series will be accepted. (0.5 pt)

8. Write a Python function without argument that returns the smallest integer j such that $P(X \leq j) \geq 0{,}95$. Run the function and check the minimality of the result. (1 pt)

Final white · Zero 1

SUBJECT ZERO · TRAINING

Duration: 4 hours · Rating of 20 · Four independent exercises.
Authorised calculator. All answers must be justified; incomplete steps will be taken in account. Give the exact values; round the probability to 10^{-6} and other approximations to 10^{-4}, unless otherwise specified.

Name: ..... **Class:** ..... **Date:** .....

Exercise 1 · A logarithmic function — 5 points

The function is considered f defined on $]0;+\infty[$ by $f(x)=\frac{2+\ln x}{x}$. We note $\mathcal{C}_f$ its representative curve.

1. Determine the limits of f in 0^+ and in $+\infty$. (0.5 pt)

2. Calculate $f'(x)$, then draw up the table of variations of f. (0.5 pt)

3. Specify the maximum value of f, then solve $f(x)=0$. (0.5 pt)

4. Study the convexity of f and determine the coordinates of the deflection point of $\mathcal{C}_f$. (1 pt)

5. Demonstrate that the equation $f(x)=\frac{1}{2}e^1$ admits exactly two solutions in $]0;+\infty[$. (0.5 pt)

6. Determine a antiderivative of f on $]0;+\infty[$. (0.5 pt)

7. Calculate the average value of f on $[1;e]$. (0.5 pt)

8. Irrespective of previous issues, resolve $\ln(x-2)+\ln(x+2)=\ln(32)$ after having specified the set on which this equation is defined. (1 pt)

Exercise 2 · Periodic phenomenon — 5 points

A point moves on a wheel rotating at constant speed in a vertical plane. $h(t)=9-3\cos\left(\frac{\pi t}{5}\right)$ where: t is expressed in seconds and $h(t)$ in above the ground is modelled by We're studying a full tour, $t\in[0;10]$.

1. Give an extended model period to $\mathbb{R}$, then the minimum and maximum heights on a turn. (0.5 pt)

2. Calculate $h'(t)$ and to draw up a table of changes in h on $[0;10]$. (0.5 pt)

3. Determine every moment $t\in[0;10]$ such that $h(t)=\frac{21}{2}$. (0.5 pt)

4. Calculate the average value of h on $[0;10]$. (1 pt)

5. Study the convexity of h the study interval and determine the coordinates of the points his curve inflection. (0.5 pt)

6. Determine the total duration during which the height is at least equal to the level of the question 3. (0.5 pt)

7. The vertical speed is given by $h'(t)$. Determine its maximum and minimum values, in metres per second, as well as the moments when they are reached. (0.5 pt)

8. Calculate vertical displacement $h(10)-h(0)$, then the sum of the height travelled at the climb explain why these two results are different. (1 pt)

Final white · Zero 1 · continued

Exercise 3 · Right, plan and planned — 5 points

In an orthonormal space coordinate system, the plane is considered $\mathcal{P}$ (d) equation $x + 2y + 2z = 11$ and point $M(2;0;0)$. The right d go through M and admits as a guiding vector $\vec{n} = (1;2;2)$.

1. Provide a parametric representation of d. (0.5 pt)

2. Demonstrate that d cut $\mathcal{P}$ in one point H, then calculate the coordinates of H. (0.5 pt)

3. Justify that H is the orthogonal project of M on $\mathcal{P}$, then calculate the distance from M to $\mathcal{P}$. (0.5 pt)

4. The plan is considered $\mathcal{Q}$ (d) equation $y = z$. Demonstrate that $\mathcal{P} \cap \mathcal{Q}$ is a right and give one parametric representation. (1 pt)

5. Verify that $H \in \mathcal{Q}$. Deducing the distance from M right $\mathcal{P} \cap \mathcal{Q}$. (0.5 pt)

6. Determine a Cartesian equation of the plane parallel to $\mathcal{P}$ passing through M. (0.5 pt)

7. Determine the angle between d and the right $\mathcal{P} \cap \mathcal{Q}$. (0.5 pt)

8. Define point N by $\overrightarrow{MN} = \vec{n}$. Calculate distances from N on the plan $\mathcal{P}$ and to the right $\mathcal{P} \cap \mathcal{Q}$, Then compare them. (1 pt)

Exercise 4 · Repetition and estimation — 5 points

We repeat 20 times a two-run event, called success and failure. The tests are independent and the probability of success is $p = \frac{1}{5}$ each test. X the number of successes obtained.

1. Justify that X follows a binomial distribution and specifies its parameters. (0.5 pt)

2. Calculate the probability of achieving exactly two successes. (0.5 pt)

3. Express with the help of contrary events, then calculate, the probability of obtaining at least two success. (0.5 pt)

4. Calculate the expectation and variance of X. (1 pt)

5. We now repeat the same test a number $N \geq 1$ of times, always independently. Determine the smallest integer N for which the probability of achieving at least one success is An array or loop search is permitted. (0.5 pt)

6. For an integer $j \in \{0, \ldots, 20\}$, demonstrate the formula of $P(X = j)$ by counting positions possible successes. (0.5 pt)

7. For N independent tests of the same parameter p, we note F_N the frequency of success. Demonstrate expressions of $E(F_N)$ and $V(F_N)$ according to the N and p. (0.5 pt)

8. Using Bienaymé-Chebychev, determine a size N sufficient to ensure $P(|F_N - p| < 0{,}02) \geq 0{,}95$. Is the size obtained necessarily the smallest possible for this probability ? Justify. (1 pt)

Final white · Zero 2

SUBJECT ZERO · TRAINING

Duration: 4 hours · Rating of 20 · Four independent exercises.
Authorised calculator. All answers must be justified; incomplete steps will be taken in account. Give the exact values; round the probability to 10^{-6} and other approximations to 10^{-4}, unless otherwise specified.

Name: ..... **Class:** ..... **Date:** .....

Exercise 1 · A logarithmic function — 5 points

The function is considered f defined on $]0;+\infty[$ by $f(x)=\frac{5+\ln x}{x}$. We note $\mathcal{C}_f$ its representative curve.

1. Determine the limits of f in 0^+ and in $+\infty$. (0.5 pt)

2. Calculate $f'(x)$, then draw up the table of variations of f. (0.5 pt)

3. Specify the maximum value of f, then solve $f(x)=0$. (0.5 pt)

4. Study the convexity of f and determine the coordinates of the deflection point of $\mathcal{C}_f$. (1 pt)

5. Demonstrate that the equation $f(x)=\frac{1}{2}e^4$ admits exactly two solutions in $]0;+\infty[$. (0.5 pt)

6. Determine a antiderivative of f on $]0;+\infty[$. (0.5 pt)

7. Calculate the average value of f on $[1;e]$. (0.5 pt)

8. Irrespective of previous issues, resolve $\ln(x-3)+\ln(x+3)=\ln(40)$ after having specified the set on which this equation is defined. (1 pt)

Exercise 2 · Periodic phenomenon — 5 points

A point moves on a wheel rotating at constant speed in a vertical plane. Its height above the ground is modelled by $h(t)=12-4\cos\left(\frac{\pi t}{4}\right)$ where: t is expressed in seconds and $h(t)$ in metres. We're studying a full tour, $t\in[0;8]$.

1. Give an extended model period to $\mathbb{R}$, then the minimum and maximum heights on a turn. (0.5 pt)

2. Calculate $h'(t)$ and to draw up a table of changes in h on $[0;8]$. (0.5 pt)

3. Determine every moment $t\in[0;8]$ such that $h(t)=14$. (0.5 pt)

4. Calculate the average value of h on $[0;8]$. (1 pt)

5. Study the convexity of h the study interval and determine the coordinates of the points his curve inflection. (0.5 pt)

6. Determine the total duration during which the height is at least equal to the level of the question 3. (0.5 pt)

7. The vertical speed is given by $h'(t)$. Determine its maximum and minimum values, in metres per second, as well as the moments when they are reached. (0.5 pt)

8. Calculate vertical displacement $h(8)-h(0)$, then the sum of the height travelled at the climb explain why these two results are different. (1 pt)

Final white · Zero 2 · continued

Exercise 3 · Right, plan and planned — 5 points

In an orthonormal space coordinate system, the plane is considered $\mathcal{P}$ (d) equation $x + 2y + 2z = 7$ and point $M(3;0;0)$. The right d go through M and admits as a guiding vector $\vec{n} = (1;2;2)$.

1. Provide a parametric representation of d. (0.5 pt)

2. Demonstrate that d cut $\mathcal{P}$ in one point H, then calculate the coordinates of H. (0.5 pt)

3. Justify that H is the orthogonal project of M on $\mathcal{P}$, then calculate the distance from M to $\mathcal{P}$. (0.5 pt)

4. The plan is considered $\mathcal{Q}$ (d) equation $y = z$. Demonstrate that $\mathcal{P} \cap \mathcal{Q}$ is a right and give one parametric representation. (1 pt)

5. Verify that $H \in \mathcal{Q}$. Deducing the distance from M right $\mathcal{P} \cap \mathcal{Q}$. (0.5 pt)

6. Determine a Cartesian equation of the plane parallel to $\mathcal{P}$ passing through M. (0.5 pt)

7. Determine the angle between d and the right $\mathcal{P} \cap \mathcal{Q}$. (0.5 pt)

8. Define point N by $\overrightarrow{MN} = \vec{n}$. Calculate distances from N on the plan $\mathcal{P}$ and to the right $\mathcal{P} \cap \mathcal{Q}$, Then compare them. (1 pt)

Exercise 4 · Repetition and estimation — 5 points

We repeat 17 times a two-run event called success and failure. The tests are independent and the probability of success is $p = \frac{3}{10}$ each test. X the number of successes obtained.

1. Justify that X follows a binomial distribution and specifies its parameters. (0.5 pt)

2. Calculate the probability of achieving exactly two successes. (0.5 pt)

3. Express with the help of contrary events, then calculate, the probability of obtaining at least two success. (0.5 pt)

4. Calculate the expectation and variance of X. (1 pt)

5. We now repeat the same test a number $N \geq 1$ of times, always independently. Determine the smallest integer N for which the probability of achieving at least one success is An array or loop search is permitted. (0.5 pt)

6. For an integer $j \in \{0, \ldots, 17\}$, demonstrate the formula of $P(X = j)$ by counting positions possible successes. (0.5 pt)

7. For N independent tests of the same parameter p, we note F_N the frequency of success. Demonstrate expressions of $E(F_N)$ and $V(F_N)$ according to the N and p. (0.5 pt)

8. Using Bienaymé-Chebychev, determine a size N sufficient to ensure $P(|F_N - p| < 0{,}02) \geq 0{,}95$. Is the size obtained necessarily the smallest possible for this probability ? Justify. (1 pt)